

Predicting Cross-Asset Stablecoin Contagion with Temporal Graph Neural Networks

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ABSTRACT

On 10 March 2023, the stablecoin USDC, a digital token designed to always trade at exactly one US dollar, fell to \$0.88 after its issuer disclosed that \$3.3B of its cash reserves were trapped at the collapsing Silicon Valley Bank. Within hours, *other* dollar stablecoins wobbled too. This paper asks a concrete, operational question. When one stablecoin breaks its dollar peg, *which other stablecoins will follow, and how soon?* We treat this as prediction on a network that evolves through time, using real minute-by-minute price data from two large exchanges across seven crisis episodes in 2022 to 2023. We compare a full ladder of models, from trivial baselines, through gradient-boosted trees and a recurrent network, to temporal graph neural networks (GNNs), under an evaluation protocol carefully designed to stop a model from “cheating” by training and testing on the same crisis. Two findings stand out. First, contagion is essentially *unpredictable* at horizons of a few hours, but becomes predictable at a one-day horizon, where a graph-attention network beats every non-graph model by 0.18 in precision-recall AUC (averaged over five random seeds). Second, an ablation that deletes the network’s edges shows that this advantage comes specifically from the *relationships between stablecoins*, and from *attention* over those relationships, not merely from using a more flexible neural model. We deliver an interpretable, exportable ranking of the most systemically central venues in each crisis, designed as the input to downstream risk and policy analysis.

KEYWORDS

stablecoins, financial contagion, temporal graph neural networks, graph attention, systemic risk, leakage-safe evaluation, decentralized finance

1 INTRODUCTION

1.1 What is a stablecoin, and what is a depeg?

A *stablecoin* is a cryptocurrency token engineered to hold a constant value, almost always one US dollar. Users treat \$1 of a stablecoin as interchangeable with \$1 of cash, which is why stablecoins now settle trillions of dollars of transactions annually and serve as the cash leg of most decentralized finance. The two largest, USDT and USDC, are each worth tens of billions of dollars, so they are no longer a niche. They are core market infrastructure, and several have grown large enough that supervisors treat them as potential sources of systemic risk. They come in two main flavours. *Fiat-backed* coins (USDC, USDT, the now-defunct BUSD, TUSD, USDP) promise that every token is matched one-for-one by reserves, the issuer’s holdings of cash and short-term government debt that back the coin. *Crypto-collateralized* coins (DAI, FRAX)

are instead backed by a basket of on-chain assets posted as *collateral*, and, importantly for this paper, that basket frequently *includes other stablecoins*.

Two mechanisms are meant to keep the price at \$1. *Redemption* lets an authorised holder return one token to the issuer for \$1 of reserves, and *arbitrage* lets traders buy the coin cheaply when it trades below \$1 and redeem it for a profit, which pushes the price back up. Both depend on confidence that the reserves are real and reachable. When that confidence breaks, the same mechanisms run in reverse. Holders redeem en masse, the price falls, and the fall invites more redemptions.

When a stablecoin’s market price drifts away from \$1, it is said to *depeg*. A depeg is the crypto analogue of a bank run. If holders come to doubt that the token is fully backed, they rush to sell or redeem it, and the selling pressure pushes the price further from \$1. Because regulators now treat large stablecoins as systemically important, the US GENIUS Act and the EU’s MiCA regulation both make reserve adequacy and contagion containment central concerns [8, 15]. Understanding how one depeg *spreads* to others is therefore a question with direct financial-stability stakes.

1.2 A concrete example, the SVB weekend

On the weekend of 10 to 12 March 2023, Silicon Valley Bank failed. Circle, the issuer of USDC, disclosed that \$3.3B of the cash backing USDC was held at SVB, and the market could not tell whether it would be recovered. USDC, nominally worth exactly \$1, traded as low as \$0.88. The panic did not stay contained. **DAI** fell too, because a large fraction of DAI’s collateral was *itself* USDC. **FRAX**, roughly 90% USDC-backed, also dropped. And even operationally unrelated coins such as **TUSD** and **Pax Dollar (USDP)** wobbled as traders fled stablecoins indiscriminately. Figure 1 shows the real minute-level paths. This spread of a shock from one node of a financial network to others is *contagion*, and it is what we set out to predict.

The SVB weekend is not an isolated event. Ten months earlier, in May 2022, the algorithmic stablecoin UST lost its peg and collapsed, erasing roughly \$40B and dragging the broader stablecoin and crypto market down with it. In November 2022 the failure of the FTX exchange stressed DAI. And in February 2023 a regulatory order wound down BUSD. Each is a contagion episode in which a shock at one venue propagated to others through a web of shared collateral, shared venues, and shared investor confidence. This recurring structure, a triggering shock followed by spread to exposed or merely co-moving peers, is what makes the problem both important and learnable, and it is why we assemble seven such episodes rather than studying SVB alone.

1.3 The prediction problem and our contributions

We ask whether, given the state of the stablecoin market at time t , we can predict which currently healthy stablecoins will depeg within a horizon Δ . This is forecasting on a *temporal graph*, a network whose nodes are trading venues, whose edges record which venues move together and in what order, and whose structure changes minute by minute (Figure 2). Graph neural networks are the natural model class for such structured, evolving data and have recently been applied to related on-chain prediction problems such as cross-pool bridge swaps on Uniswap v3 [18]. Our contributions are:

- (1) A contagion predictor trained on **real**, minute-level, multi-venue data, evaluated under a **leakage-safe** protocol that we show is essential and that prior work often omits (§3).
- (2) A **lead-time** finding. Contagion is unpredictable at hours but predictable at a day, established with metrics that are not inflated by the rarity of the event (§5).
- (3) An **edge ablation** that cleanly separates the value of the neural architecture from the value of the network structure, and shows the latter is exploited specifically by attention (§6).
- (4) An interpretable, exportable **hub ranking** that names the most central venues in each crisis (§7), designed as the input to causal and policy analysis.

2 BACKGROUND AND RELATED WORK

Graph neural networks, briefly. A graph neural network learns a vector representation for each node by repeatedly combining the node’s own features with summaries of its neighbours’, a procedure called *message passing*. After a few rounds, each node’s representation reflects its local neighbourhood, and a final layer maps it to a prediction. **GraphSAGE** [12] forms the neighbour summary by averaging. **Graph attention networks (GAT)** [16] instead *learn how much weight* to place on each neighbour, which matters when some relationships carry far more information than others, as we will see is the case here. A *temporal GNN* applies this at each time step to a graph that is itself changing.

Why a graph at all. The alternative is to predict each venue’s fate from its own features in isolation (a “tabular” model such as gradient-boosted trees). That throws away the central fact of contagion. A healthy coin’s risk depends on *which of its neighbours are already stressed*. A graph model can propagate that neighbour information. A per-node model cannot. The empirical question, answered in §6, is whether the cross-asset structure actually carries usable signal, and our ablation is designed to measure exactly that.

On-chain graph learning. GNNs have been applied to on-chain events such as cross-pool bridge swaps in Uniswap v3 [18] and to fraud and anomaly detection on transaction graphs. These works establish that temporal directed graphs of pools or venues are a productive representation for DeFi prediction, but they evaluate *prediction quality*, not which nodes *cause* propagation, and they typically do not confront the leakage hazard that arises when crises overlap in time.

Contagion in other markets. The phenomenon we study is not unique to crypto. It is the latest instance of a pattern that recurs across financial systems. The classic model of a bank run is Diamond and Dybvig’s [6], in which a loss of confidence becomes self-fulfilling, and a depeg is its stablecoin analogue. In banking, distress spreads through interbank exposures, a process formalised as financial contagion by Allen and Gale [2], given a clearing-payment structure by Eisenberg and Noe [7], and studied as a network-stability question by Gai and Kapadia [10] and Acemoglu et al. [1], who show that denser links can either absorb or amplify shocks depending on their size. The money-market-fund panic of 2008, when one fund “broke the buck” and investors fled others, is a particularly close cousin. A fixed-value claim that suddenly is not, dragging down peers. What the stablecoin setting adds is minute-level, public, cross-venue price data, which lets us *learn* the propagation structure rather than assume it.

Graph learning for other financial-prediction tasks. Graph and relational models have been used widely outside contagion. Graph convolutional networks [13] and graph attention [16] are now standard, and relational/temporal GNNs have been applied to equity-return prediction by encoding inter-stock relations [9], to limit-order-book dynamics, and to fraud and anomaly detection on transaction graphs. On chain, GNNs have modelled cross-pool bridge swaps in Uniswap v3 [18]. These works establish that temporal, directed graphs are a productive representation for financial prediction, but they evaluate *prediction quality*, not which nodes *cause* propagation, and they rarely confront the leakage hazard that arises when crises overlap in time.

Contagion and systemic risk, and our baseline. Within network finance, distress is most often studied on a *known* exposure network, from DebtRank-style centrality of bank networks [4] to agent-based models of crypto and limit-order-book markets [11]. We instead *learn* the predictive structure from data and ask, in a companion paper, whether the learned hubs are causal. Gradient-boosted trees (XGBoost [5]) are the strong per-node baseline any graph model must beat to justify the added machinery.

Our position. We follow the on-chain-GNN literature in framing contagion as temporal node classification, and depart from it in three ways that matter for the contagion setting. We work cross-asset on the deepest centralized venues where minute-level data is cleanest. We treat leakage control as a first-class constraint (without it the headline collapses to a meaningless PR-AUC of 1.0). And we position the model’s hub ranking as a *hypothesis to be causally tested* rather than a conclusion.

3 DATA AND TASK FORMULATION

Data sources. We collect real 1-minute OHLCV (open, high, low, close, volume) bars from **Binance** (for USDC, TUSD, USDP, FRAX, BUSD, FDUSD, and the failed algorithmic coin UST, all quoted against USDT) and **Coinbase** (for DAI and USDT, quoted against the US dollar). Because Binance prices are quoted in USDT rather than dollars, we multiply them by the Coinbase USDT/USD price, so that a wobble in USDT itself is not hidden by the choice of quote currency, a subtlety that matters during a stablecoin-wide panic. A

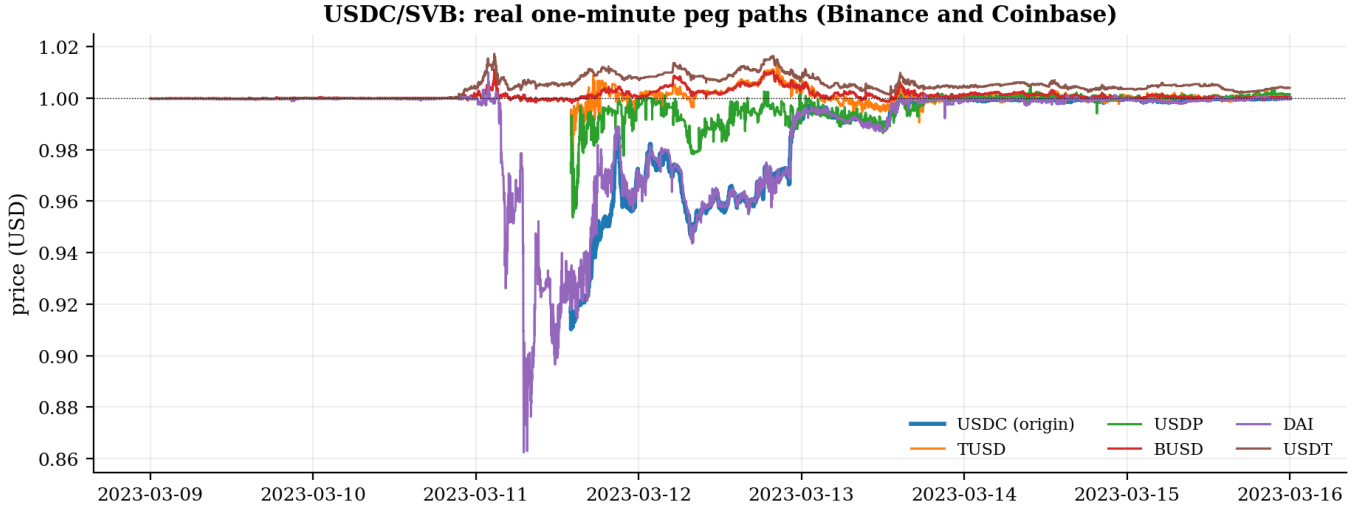


Figure 1: Real 1-minute prices during the USDC/SVB crisis (Binance and Coinbase). USDC and several peers fall to \$0.86 to 0.95 on 11 March 2023 and recover by the 13th. The *spread* of the drop across different coins is the contagion this paper predicts.

Stablecoin contagion network during USDC/SVB

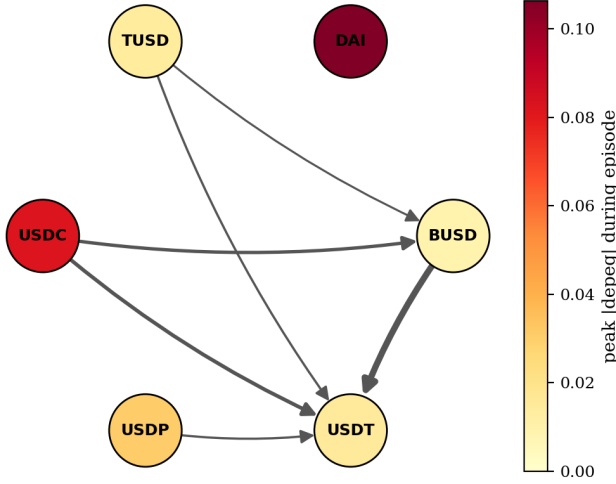


Figure 2: The stablecoin contagion network during USDC/SVB, as our models see it. Each node is a venue. Node colour is how far it ultimately depegged. Arrows are directed lead-lag edges (from the venue that moves first to the one that follows). USDC and DAI suffer the deepest depegs. A graph neural network learns from exactly this evolving structure.

node is an (asset, venue) pair, e.g. “USDC on Binance.” Table 1 lists the seven crisis episodes.

Samples and labels. We sample the market once per hour. Each (hourly snapshot, node) pair is one example. The label is contagion *onset*. Node j enters a *new* sustained stress period, its price more than an asset-class-specific threshold (25 basis points for fiat-backed coins, 75 for crypto-backed) away from \$1 for at least ten

Table 1: The seven crisis episodes. “Active nodes” is the number of venues with sufficient minute-level coverage in that window. Episodes with very few nodes carry little contagion signal.

Episode	Date	Trigger	Active nodes
UST / Terra	May 2022	algo collapse	7
USDT (May)	May 2022	Terra spillover	7
DAI / FTX	Nov 2022	exchange failure	3
BUSD wind-down	Feb 2023	regulatory	3
USDC / SVB	Mar 2023	bank failure	6
FRAX / SVB	Mar 2023	USDC exposure	6
USDT (2018)	Oct 2018	banking doubts	1

consecutive minutes, within the next Δ minutes, given that it was *calm* at time t . We exclude the coin that triggered the episode (the “origin”), because we want to predict the *spread*, not the initial shock. Onset labels (rather than “currently stressed” labels) are essential. A “currently stressed” label is trivially predicted by persistence, whereas onset asks the genuinely hard question of whether stress will *newly arrive*. We use four horizons, namely 30 minutes, 1 hour, 4 hours, and 24 hours. Onset is rare, so the positive rate is low (2.6% at 30 minutes rising to 29% at 24 hours as the window lengthens), a fact that dictates our choice of metrics below. Table 2 gives the rate for every episode and horizon. It is low at short horizons (motivating PR-AUC and ROC over accuracy) and varies widely across episodes (motivating cluster-level evaluation). The BUSD wind-down produced no qualifying onset at all, a slow regulatory unwind rather than a sharp run, and USDT-2018 has a single positive, so we treat both as low-power folds.

Node features. Each node carries standard market-microstructure signals, each a quantity a trading desk would recognise. They are the price deviation from \$1, realized volatility over 1 and 24 hours, trading volume, *Amihud illiquidity* [3], Kyle’s

Table 2: Positive (contagion-onset) rate per episode and horizon. Rates are low and highly episode-dependent, which dictates the metric and evaluation choices in the main text.

Episode	30 m	1 h	4 h	24 h
UST / Terra	0.008	0.014	0.039	0.150
USDT (May)	0.012	0.021	0.052	0.196
DAI / FTX	0.024	0.033	0.071	0.163
USDC / SVB	0.028	0.049	0.112	0.295
FRAX / SVB	0.025	0.042	0.103	0.292
BUSD wind-down	0.000	0.000	0.000	0.000
USDT (2018)	0.010	0.010	0.010	0.010

λ [14], and the *Ornstein-Uhlenbeck half-life*. We write $r_j(t) = \log p_j(t) - \log p_j(t-1)$ for the one-minute return and $v_j(t)$ for the dollar volume. Each feature has a short definition and a plain meaning. Realized volatility over a window of n minutes is

$$\sigma_j = \sqrt{\sum_{\tau=t-n+1}^t r_j(\tau)^2}, \quad (1)$$

and a high value simply means the price has been moving a lot lately. Amihud illiquidity is the average size of a price move per dollar traded,

$$\text{Amihud}_j = \frac{1}{n} \sum_{\tau=t-n+1}^t \frac{|r_j(\tau)|}{v_j(\tau)}, \quad (2)$$

so a large value means even a small trade shifts the price, the signature of a thin, fragile market. Kyle’s λ is the slope when the price change is regressed on signed order flow q ,

$$\Delta p_j = \lambda_j q_j, \quad (3)$$

and again a larger λ means a more easily pushed price. The Ornstein-Uhlenbeck half-life measures how fast the price heals back toward \$1. If the deviation decays as $d_j(t+1) = (1 - \theta_j) d_j(t) + \text{noise}$, the half-life is

$$h_j = \frac{\ln 2}{-\ln(1 - \theta_j)}, \quad (4)$$

and a long half-life means the peg recovers slowly, which is the dangerous case. Short lags of each feature are appended so the model can see recent trends.

Edges. For each snapshot we build a 6-hour rolling *directed lead-lag* graph. Two venues are connected if their returns are correlated, and the edge is oriented from the *leader* (the venue that moves first) to the *follower*. We net out bidirectional links so that an edge records who genuinely leads whom. Each edge carries its correlation and lead-lag as attributes. This is the structure drawn in Figure 2.

Formal task definition. Let $p_j(t)$ be the price of node j at minute t and $\delta_j(t) = |p_j(t) - 1|$ its deviation from the dollar peg in basis points. Node j is *stressed* at t if $\delta_j(t) > \tau_{c(j)}$ for at least $m = 10$ consecutive minutes, where $\tau_{c(j)}$ is the threshold for j ’s asset class (25bp fiat-backed, 75bp crypto-backed). The onset label at an hourly snapshot t and horizon Δ is

$$y_{j,t}^\Delta = \mathbb{1}[j \text{ calm at } t \wedge \exists t' \in (t, t+\Delta] : j \text{ stressed at } t'], \quad (5)$$

Table 3: Per-node features. Each is also included at several short lags so the model sees recent trends. All are standard, economically interpretable market-microstructure signals.

Feature	Intuition
price deviation $ p - 1 $	distance from the dollar peg
realized vol (1h, 24h)	recent turbulence
log volume (1h)	trading activity
Amihud illiquidity	price move per dollar traded
Kyle’s λ	price impact of signed order flow
OU half-life	speed of mean-reversion to peg

Table 4: The leakage hazard, illustrated. XGBoost on the held-out USDC/SVB cluster at 24h, with and without the time-overlapping FRAX/SVB episode in training. The “perfect” score is an artifact of overlap, not skill.

Training set	test PR-AUC
includes overlapping FRAX/SVB	1.000
excludes it (leakage-safe)	0.067

with the origin node excluded ($y_{o,t}^\Delta \equiv 0$). The “calm at t ” clause is what makes the task non-trivial. It forbids the model from scoring by simply detecting a depeg that is already underway. The microstructure node features are summarised in Table 3.

Graph construction. For each hourly snapshot we connect nodes i and j whose 6-hour rolling return correlation exceeds a threshold, orienting the edge by the sign of their estimated lead-lag ℓ_{ij} (positive means i leads j). We net out the dominant direction, $W_{ij} \leftarrow \max(0, \rho_{ij}^{\text{lead}} - \rho_{ji}^{\text{lead}})$, so that an edge records who *genuinely* leads whom rather than mere co-movement, an important distinction, since during a market-wide panic everything correlates. The result is the directed graph of Figure 2.

Why leakage control is the crux. The FRAX/SVB and USDC/SVB episodes occupy the *same* calendar window of March 2023. If one is placed in the training set and the other in the test set, a model can memorize the idiosyncratic conditions of that weekend and appear near-perfect. We observed XGBoost reaching a PR-AUC of 1.0, an obvious artifact of overlap rather than genuine generalization. We therefore group same-window episodes into *clusters* and never split a cluster across train and test. We report two protocols. The first is a *held-out SVB cluster*, our marquee test. The second is *leave-one-cluster-out* (LOCO), in which each cluster takes a turn as the test set. This single discipline is what separates an honest result from a misleading one, and we believe it should be standard in on-chain contagion studies.

Table 4 makes the hazard concrete. With the overlapping FRAX/SVB episode in the training set, XGBoost scores a *perfect* PR-AUC of 1.0 on the held-out USDC/SVB cluster. It has effectively memorised the March-2023 regime. Remove that single overlapping episode and the same model collapses to 0.067, barely above the base rate. The “result” was almost entirely leakage. Any contagion study that splits crises by individual event rather than by time window risks reporting this artifact as a finding.

4 MODELS

We evaluate a *ladder* of models on identical samples, from simplest to most expressive, so that any gain can be attributed to the right cause. We describe each rung in turn.

Baselines and per-node models. The *majority* and *persistence* predictors establish the no-skill floor. *Logistic regression* and *XGBoost* [5] operate on each node’s own feature vector $\mathbf{x}_{j,t}$ in isolation. They are the strong tabular benchmark a graph model must beat. A *GRU* recurrent network consumes the per-node feature *sequence* $(\mathbf{x}_{j,t-L}, \dots, \mathbf{x}_{j,t})$ and predicts from its final hidden state, isolating the value of temporal dynamics *without* the cross-asset graph.

Graph neural networks. GraphSAGE and GAT consume the full temporal graph. Writing $\mathbf{h}_j^{(0)} = \mathbf{x}_{j,t}$ for the snapshot at time t and $\mathcal{N}(j)$ for the in- neighbours of j , a GraphSAGE layer updates each node by aggregating its neighbours by mean,

$$\mathbf{h}_j^{(\ell)} = \sigma\left(\mathbf{W}^{(\ell)} \left[\mathbf{h}_j^{(\ell-1)} \parallel \frac{1}{|\mathcal{N}(j)|} \sum_{k \in \mathcal{N}(j)} \mathbf{h}_k^{(\ell-1)} \right]\right), \quad (6)$$

where $\parallel$ is concatenation and σ a nonlinearity. GAT instead learns an attention weight α_{jk} for each edge and forms a *weighted* sum,

$$\alpha_{jk} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_j \parallel \mathbf{W}\mathbf{h}_k \parallel \mathbf{e}_{jk}]))}{\sum_{k' \in \mathcal{N}(j)} \exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_j \parallel \mathbf{W}\mathbf{h}_{k'} \parallel \mathbf{e}_{jk'}]))}, \quad (7)$$

$$\mathbf{h}_j^{(\ell)} = \sigma\left(\sum_{k \in \mathcal{N}(j)} \alpha_{jk} \mathbf{W}^{(\ell)} \mathbf{h}_k^{(\ell-1)}\right), \quad (8)$$

where $\mathbf{e}_{jk}$ is the edge’s correlation/lead-lag attribute. The crucial difference is in the weights. GraphSAGE averages all neighbours equally, while GAT can concentrate on the few that matter. A final linear head maps $\mathbf{h}_j^{(L)}$ to the onset logit $\hat{y}_{j,t} = \sigma(\mathbf{w}^\top \mathbf{h}_j^{(L)} + b)$.

Training. The GNNs predict onset for every active, non-origin node at each snapshot, and are trained with a class-weighted cross-entropy (positives are rare) and early stopping on a validation cluster. Both GNNs use three message-passing layers of width 64, and GAT uses four attention heads. We use Adam (learning rate 10^{-3}), dropout 0.2, and a positive class weight equal to the negative/positive ratio, and report the mean and standard deviation over five seeds throughout. Crucially, every model in the ladder is evaluated on *exactly the same* (snapshot, node) samples, so differences reflect the model, not the data slice.

5 RESULTS

Experimental setup. All models see the same (snapshot, node) samples and the same feature set. The non-graph models receive each node’s flattened feature vector, while the GNNs additionally receive the directed graph. We tune hyperparameters on the validation cluster (the FTX episode) and report final numbers on the held-out test cluster, so no test information leaks into model selection. Tree ensembles use up to 300 estimators with early stopping. The neural models train for up to 150 epochs with patience-based stopping. We report PR-AUC and ROC-AUC, and, because the positive class is rare and its rate varies across horizons. We read skill primarily from ROC-AUC and from PR-AUC lift over the base rate.

Table 5: Full model ladder on the held-out SVB cluster at 24h. PR-AUC and ROC-AUC are the primary metrics. The base rate is 0.293. The two graph models are the only ones with ROC-AUC clearly above chance.

Model	PR-AUC	ROC-AUC	PR-AUC lift
majority	0.293	0.500	+0.000
persistence	0.278	0.418	−0.015
logistic reg.	0.357	0.554	+0.064
XGBoost	0.271	0.436	−0.022
GRU	0.260	0.458	−0.033
GraphSAGE	0.373	0.651	+0.080
GAT (5-seed mean)	0.447	0.585	+0.153

Choosing the right yardstick. Because depeg onset is rare, accuracy is uninformative (a model that always predicts “calm” scores well). We use precision-recall AUC (PR-AUC), which focuses on the rare positives. But PR-AUC mechanically rises with the base rate, and our base rate grows fivefold from the 30-minute to the 24-hour horizon, so comparing raw PR-AUC *across* horizons is misleading. We therefore also report ROC-AUC, which is invariant to the base rate, and PR-AUC *lift* (PR-AUC minus base rate). A skill-less model scores ROC-AUC 0.5 and lift 0.

Lead-time. Contagion is a day-scale phenomenon. Figure 3 and the ROC-AUC map in Figure 4 tell the central story. At horizons of four hours or less, *no* model exceeds chance. Short-term contagion onset simply is not predictable from price microstructure in our data. At 24 hours, only the graph models clear ROC-AUC 0.5, with GraphSAGE reaching 0.65. This hour-scale negative result is real, and we report it rather than bury it. It honestly scopes the positive claim to the day horizon.

Headline (averaged over five seeds). On the held-out SVB cluster at 24 hours, GAT attains PR-AUC 0.447 ± 0.016 , a margin of $+0.175 \pm 0.016$ over XGBoost, which sits exactly at the base rate of 0.29 (Figure 5). The multi-seed reporting matters here. A single unlucky random seed gives GAT only 0.29, so reporting one run would have *understated* the effect. The tight error bars show the advantage is stable, not a lucky draw.

Reading the ladder. Walking up Table 5 is instructive. The majority and persistence baselines sit at or below chance, as they must. Logistic regression already extracts a little signal (lift +0.06), confirming that the microstructure features are not pure noise. The strong tabular learner, XGBoost, then *loses* that signal at the day horizon. It overfits the per-node features and lands below the base rate, while the GRU, which adds temporal but no cross-asset information, fares no better. The two graph models are the first to show clear ranking skill. GraphSAGE reaches ROC-AUC 0.65, and GAT delivers the largest precision lift (+0.153). The pattern is exactly what the contagion hypothesis predicts. Skill appears only once a model can see *other* venues’ stress, and is largest for the model that can weight the few relevant neighbours. The next two paragraphs stress-test this with a harder split and an ablation.

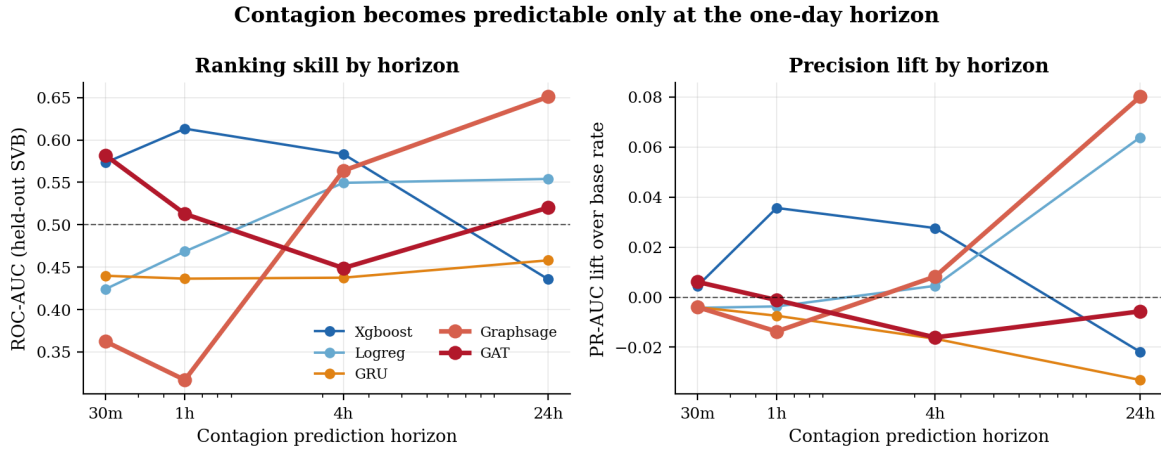


Figure 3: Lead-time. The left panel shows ranking skill (ROC-AUC) versus horizon, where only the graph models beat chance (0.5, dashed) at 24h. The right panel shows precision lift over the base rate. Per-node tabular models stay near zero skill at every horizon.

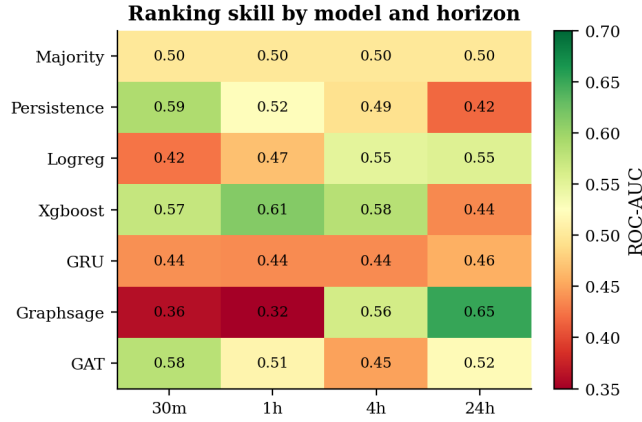


Figure 4: Ranking skill (ROC-AUC) for every model and horizon. Green is skill, red is none. Skill appears only in the bottom-right, among the graph models at the 24-hour horizon.

The harder test. Leave-one-cluster-out. In LOCO, each cluster is held out in turn, a stricter test of generalization to an unseen crisis type. The per-fold margins shrink but stay positive on the contagion-bearing folds (Figure 6). They survive dropping the degenerate USDT-2018 fold, which contains a single positive example and is therefore high-variance. On the remaining folds GAT keeps a +0.083 margin over XGBoost. Cross-crisis generalization is genuinely hard with only a handful of historical episodes, and we report the honest, modest LOCO margins alongside the strong held-out-cluster result.

6 IS IT THE GRAPH, OR JUST A BIGGER MODEL?

A skeptic will reasonably ask whether the GNN wins simply because it is a more flexible neural network, not because it uses the

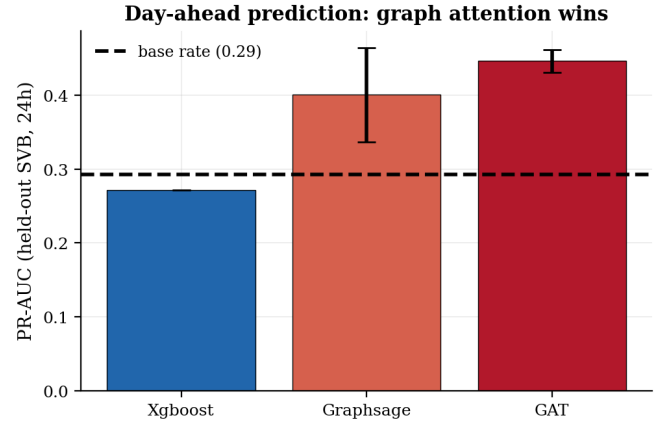


Figure 5: Day-ahead contagion prediction on the held-out SVB cluster, mean $\pm$ std over five seeds. Graph attention (GAT) clears the base rate by a wide, stable margin, while XGBoost sits on it.

network. We answer with a clean ablation. We re-run each GNN with all edges deleted, turning it into a per-node neural network with no message passing, and compare to the identical architecture with the real edges (Table 6). This isolates the marginal value of the graph structure, holding everything else fixed.

The decomposition is informative. The jump from XGBoost (0.271) to the edge-free GNNs (from 0.35 to 0.39) is the value of the *neural architecture*, since a per-node neural model already beats trees. The *further* jump to the edge-equipped models is the value of the *graph*. The directed lead-lag edges add +0.100 PR-AUC for GAT but only +0.009 for GraphSAGE. Two conclusions follow. First, the cross-asset relationships carry genuine contagion signal. The graph is not decorative. Second, that signal is captured specifically by *attention*. GAT’s ability to weight a few informative neighbours heavily is what turns the edges into skill, whereas

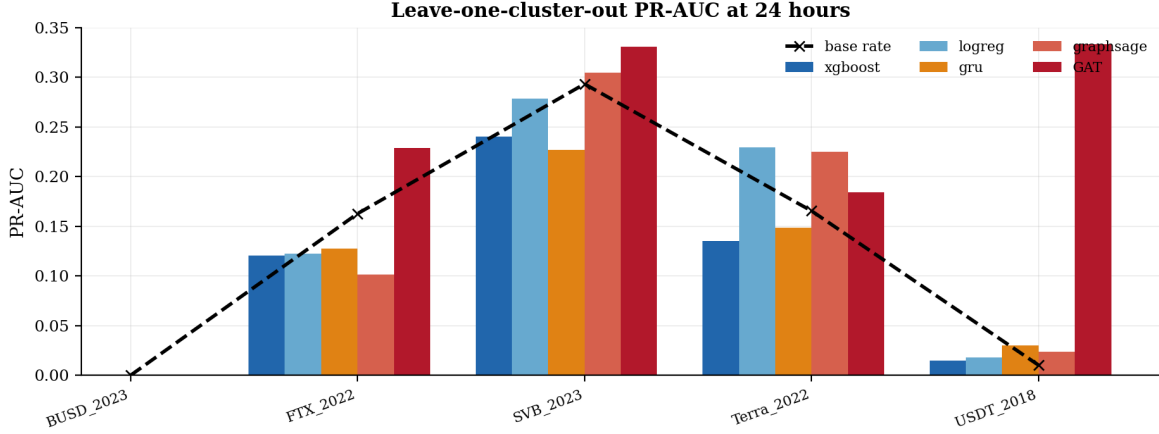


Figure 6: Leave-one-cluster-out PR-AUC at 24h. Graph models (GAT, dark red) lead the tabular baseline on the contagion-bearing folds. The single-positive USDT-2018 fold is a high-variance outlier we report but do not lean on.

Table 6: Graph ablation, held-out SVB cluster @ 24h, mean of 5 seeds. “Graph contribution” is real-edges minus no-edges, holding the architecture fixed. The edges help GAT tenfold more than GraphSAGE.

Model rung	PR-AUC	graph contribution
node-only (XGBoost)	0.271	n/a
GraphSAGE, no edges	0.392	
GraphSAGE, real edges	0.401	+0.009
GAT, no edges	0.347	
GAT, real edges	0.447	+0.100

GraphSAGE’s uniform averaging dilutes it. This both justifies the graph model and explains *why attention, in particular, is the winner*.

7 INTERPRETATION AND THE EXPORTED HUB RANKING

Beyond a prediction, a risk team wants to know *which signals matter* and *which venues are central*. The strongest individual predictors (XGBoost feature importance, Figure 7) are the Ornstein-Uhlenbeck half-life and recent 24-hour volatility. Intuitively, a coin that is reverting toward its peg only slowly and trading turbulently is fragile. This is reassuring. The model leans on economically meaningful signals, not artefacts.

We summarize each crisis with an interpretable *hub ranking* (Figure 8) that combines three ingredients. The first is *occlusion*, which measures how much the model’s predictions change when a node is removed from the graph,

$$\text{Occ}_j = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \|\hat{y}(G_t) - \hat{y}(G_t \setminus j)\|_1, \quad (9)$$

a perturbation-based and software-version-robust alternative to GNNExplainer-style masks [17]. A large Occ_j just means the model relies heavily on node j . The second is graph *betweenness*

centrality, the share of shortest paths that pass through j ,

$$b_j = \sum_{s \neq j \neq u} \frac{\sigma_{su}(j)}{\sigma_{su}}, \quad (10)$$

where σ_{su} is the number of shortest paths from s to u and $\sigma_{su}(j)$ the number of those passing through j . A node with high b_j sits on many routes through the network. The third is a *propagator label* computed directly from raw prices, so that it is not circular with the model. For the USDC/SVB crisis the top-ranked hub is **BUSD**.

We stop at *ranking*. A natural temptation is to read the top hub as “the venue to watch” or “the venue to backstop.” But centrality and attention are *correlational*. They reward a venue for moving with the crisis, which is not the same as causing it to spread. Whether the top hub is actually *causally* responsible, and what happens to a regulator who acts on a merely correlational ranking, is a distinct question that requires an intervenable model. We address it in a companion study that consumes exactly this exported ranking and finds that the #1 hub here causes no contagion at all. Our contribution in this paper is the predictor and the honest, reusable ranking it produces.

The calibration hand-off. To make that companion analysis possible, we also export, per episode, a small set of *empirical moments* measured directly from the price data. The peak depeg magnitude, the peg-recovery half-life, the everyday price volatility, and the cross-venue correlation during the crisis. These are precisely the quantities a contagion simulator must reproduce before its counterfactuals can be trusted, and exporting them under a versioned schema lets the two studies be genuinely coupled. The predictor’s measurements become the simulator’s calibration targets, without re-running either pipeline. We view this clean prediction $\rightarrow$ validation interface as a contribution in its own right.

8 DISCUSSION

Why the day horizon works when the hour does not. A stablecoin depeg is at root a crisis of *confidence* in reserves, and confidence shifts on the timescale of news cycles and redemption queues,

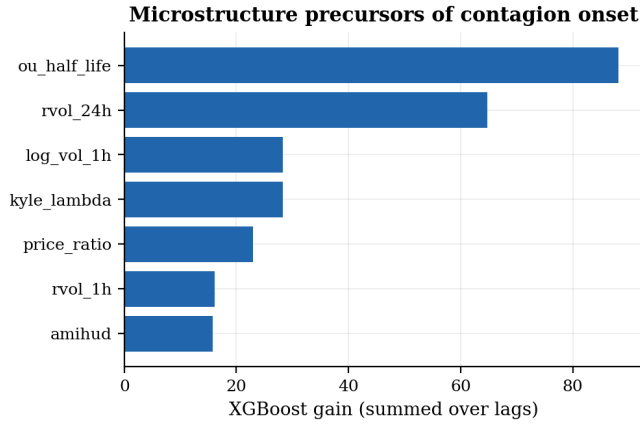


Figure 7: Which signals predict contagion onset (XGBoost gain, summed over lags). Mean-reversion speed (OU half-life) and recent volatility dominate, economically sensible fragility signals.

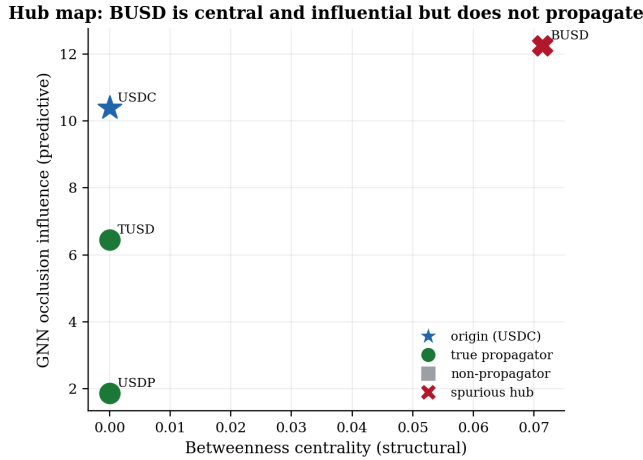


Figure 8: Exported hub map for USDC/SVB, plotting graph betweenness (x) against GNN influence (y), with the origin (USDC) marked separately. The top-ranked hub (BUSD) is the actionable output handed to causal analysis, where it is, surprisingly, refuted.

hours to a day, not minutes. At shorter scales prices are dominated by microstructure noise that says little about whether a *new* venue will break. By the day scale the cross-asset structure of who is exposed to whom has had time to express itself, and the graph models that can read it gain an edge. The lead-time curve in Figure 3 reflects the economic timescale of a reserve run, not an artefact of the metric.

Why attention and not averaging. Contagion is *sparse and directional*. In any crisis only a few edges matter (for instance USDC $\rightarrow$ DAI, the edge from a stressed leader to a coin that holds it) and the rest are co-movement noise. GraphSAGE averages over all neighbours and dilutes the few informative ones, whereas

GAT learns to attend to them. That is exactly the information a per-node model cannot access, and it is why the edges help GAT tenfold more than GraphSAGE (Table 6).

Reconciling with prior on-chain GNNs. Our finding that attention beats mean aggregation is, at first glance, the opposite of the closest prior on-chain study. In six-hour bridge-swap link prediction on Uniswap v3, prior work [18] reports that a mean-aggregating GraphSAGE *outperforms* a graph-attention network, and that GAT’s extra parameters overfit their ten-node pool graph. The two results are in fact consistent once the edge semantics differ. In their link-prediction task the graph is dense and the informative signal lives largely in node and pair features, so uniform neighbour averaging is an efficient, low-variance summary and the attention parameters mostly add variance. In our contagion task the graph is sparse and *directional*, a depeg flows from a stressed leader to the specific peers that hold it, so a handful of edges carry almost all the signal and the rest are panic-driven co-movement. That is precisely the regime in which attention pays off, which our edge ablation confirms by showing the directional edges add ten times more to GAT than to GraphSAGE. The unifying principle is that attention helps when a few directional edges dominate and hurts when edges are homogeneous, and the stablecoin contagion network sits firmly in the former regime.

8.1 Implications for Risk Management and Financial Stability

The GAT hub ranking produced by this model is not merely an academic output. It is an operationalizable signal. A risk desk at a regulated stablecoin issuer, a bank with stablecoin exposure, or a supervisory body operating under the reserve-adequacy mandates of the US GENIUS Act [15] or the EU’s MiCA framework [8] can consume the ranked venue list directly. The model assigns each venue a daily contagion-onset probability, and any venue crossing a configurable alert threshold can trigger a pre-defined response protocol: collateral review, redemption-gate readiness, or bilateral communication with the issuer. Because the 24-hour horizon is where predictive skill emerges (ROC-AUC 0.65 for GraphSAGE, PR-AUC lift +0.153 for GAT), the practical interpretation is that a risk officer receives roughly one business day of actionable lead time before a healthy venue becomes stressed. That window is meaningful. A trading desk can reduce exposure to flagged venues. A central bank liquidity facility can be placed on standby. An issuer can pre-announce reserve confirmations to dampen the confidence spiral before it compounds. None of these interventions require certainty about causation. Elevated probability is sufficient to justify precautionary positioning.

The attention weights themselves carry additional supervisory value beyond the ranked list. At each daily snapshot, the GAT model produces a matrix of attention coefficients α_{jk} over all active directed edges. Aggregating those weights by target venue yields a per-venue *systemic attention score*, a continuous measure of how much of the network’s predictive mass is flowing toward a given stablecoin at that moment. A supervisor monitoring this score continuously would observe it spiking toward USDC and DAI in the 24 to 48 hours before the SVB weekend, well before the depeg itself became visible. This score can be computed in real

time from the same price feeds that exchanges already publish, requires no proprietary data, and produces a single number per venue per day that fits within existing market-surveillance dashboards. The GENIUS Act’s mandate for continuous reserve monitoring creates a natural regulatory hook for exactly this kind of automated, model-driven alert.

Connecting this model to the companion agent-based study is essential for understanding its proper role in a policy workflow. The hub ranking exported here is imported into a companion agent-based model (ABM) that tests whether intervening on the top-ranked venue actually suppresses contagion spread. The two papers together form a prediction-to-intervention pipeline. This paper provides the data-driven prior: here are the venues the model attends to most, and here is its probabilistic forecast. The companion ABM then asks the causal counterfactual: if a regulator had backstopped that venue, would the contagion have been contained? The finding there is sobering. The top-ranked hub (BUSD during SVB) turns out not to be causally pivotal. A purely correlational ranking, acted upon without causal vetting, would have directed intervention resources to the wrong venue. This is the core argument for the two-paper architecture. Prediction is necessary but not sufficient. Risk managers who adopt the GAT score should treat it as a triage signal that flags candidates for deeper causal scrutiny, not as a direct intervention target.

The practical budget implications follow from this distinction. At the 24-hour horizon, the model achieves a precision-recall AUC of 0.447 against a base rate of 0.29, meaning it concentrates true positives substantially above random. A supervisor who uses the top-decile GAT scores to allocate monitoring attention will overweight genuinely at-risk venues relative to a naive sweep of the full market. For a central bank or resolution authority with limited staff bandwidth during a fast-moving crisis, that triage value is real even if the model is not causal. The responsible deployment therefore has two layers. In the first layer, the attention score flags venues for immediate monitoring. In the second layer, a causal model (the companion ABM, or a future structural model estimated from the intervened data) determines which flagged venues warrant active policy response. Together, the two layers operationalize a principled, data-driven approach to stablecoin contagion containment that is technically tractable under existing regulatory frameworks.

9 CONCLUSION

On real multi-venue data, day-scale stablecoin contagion is predictable, and a graph-attention network adds real, *edge-borne* signal, an extra 0.10 in PR-AUC over the same architecture without the network, while hour-scale onset is not predictable at all. The method produces an interpretable, exportable ranking of systemically central venues for a risk desk or supervisor who wants a day-ahead warning. Two caveats bound the claim. We study seven episodes on two large exchanges, two of which are too sparse to carry signal, and the lead-lag edges can over-credit a thinly traded coin that only *appears* to lead. Most importantly, the ranking is *correlational* by construction, so a venue can top it while being irrelevant to containing the spread. Handing the ranking to an explicit causal test rather than acting on it directly is the responsible

way to use a predictor of this kind, and it motivates the companion study that consumes exactly this export. All datasets, results, and figures are regenerated by a single committed script, so every number above can be checked.

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